# vol-eval: A Python Package for Volatility Forecast Evaluation, with a Re-Analysis of 30 Published Comparison Papers (2018–2025)

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**Abstract**

This paper introduces `vol-eval`, an open-source Python package that ships the canonical battery of significance tests for volatility forecast evaluation. The battery covers QLIKE, MSE, MAE, Mincer-Zarnowitz $R^2$, Diebold-Mariano (1995) with Newey-West HAC standard errors, the Hansen-Lunde-Nason (2011) Model Confidence Set with stationary bootstrap, Hansen's (2005) Superior Predictive Ability test with three p-value variants, and White's (2000) Reality Check. The package follows a functional API with typed result objects, ships under MIT license, and is verified by 34 unit tests including cross-validation against the `arch` package's reference implementation.

We apply `vol-eval` in two settings. The first is a self-contained seven-model horse race on S&P 500 realized volatility, using the panel of Khan (2026c). The second is a sample of 30 published volatility-forecasting comparison papers from 2018–2025, drawn from the *Journal of Financial Econometrics*, three open-access MDPI venues, and arXiv q-fin.

Three layered findings emerge. First, a disclosure crisis. The strict-reading mean Reproducibility Disclosure Score across the 30-paper sample is 0.47, materially lower than the 0.78 reported by Khan (2026b) for the broader 99-paper ML-finance sample. Zero of the 30 papers ship both code and publicly accessible data. Second, a methodological gap. 27 of 30 papers (90%) declare a headline winner without applying any formal predictive-accuracy test (DM, MCS, SPA, or

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Reality Check). Third, a robustness gap. In the 14 papers we could reproduce (the strict-RDS-1 subset, where data is publicly accessible), 12 paper-claimed wins (86%) do not survive significance testing under the `vol-eval` battery. Only one paper has its claim survive in full, with one additional paper showing partial survival on a subset of assets. Each per-paper reproduction and its caveats are documented in detail.

The full `vol-eval` source, the per-paper reproductions, and an audit script that re-derives every numerical claim in this paper from JSON outputs are released at github.com/ayk5511/vol-eval, scoring 2 on the Reproducibility Disclosure Score rubric of Khan (2026b).

**Keywords:** volatility forecasting, Diebold-Mariano test, Model Confidence Set, Hansen SPA, Reality Check, QLIKE, forecast evaluation, reproducibility, open-source software

**JEL Classification:** C12, C15, C22, C52, C53, G17

# Contents

# 1 Introduction

The standard practice in volatility-forecasting research is to compute a loss function on each of several candidate forecasters, rank them, and report the lowest-loss model as the winner. This practice has been the convention for at least three decades. It is also, in a meaningful fraction of published comparisons, statistically indefensible. The headline gap between the winner and runner-up is often smaller than the standard error of the loss differential; the "win" that appears decisive in a clean table does not survive a basic Diebold-Mariano test.

The methodological tools to do this correctly have existed since Diebold and Mariano (1995). Refinements that handle multiple comparisons (Hansen et al., 2011), that test a benchmark against a set of competitors (Hansen, 2005), and that protect against data-snooping bias when many forecasters are tried (White, 2000) are all between fifteen and thirty years old. The tools are not new. What is missing in the practice of the volatility-forecasting subfield is the discipline of applying them by default.

We make two contributions. First, we introduce `vol-eval`, an open-source Python package that bundles the standard loss functions and significance tests behind a clean, functional API designed to make rigorous evaluation a one-line addition to any existing forecasting workflow. The package is MIT-licensed, has 34 unit tests, and is cross-validated for numerical correctness against the canonical `arch` package (Sheppard et al., 2023). Second, we demonstrate the package by applying it to the seven-model volatility-forecasting horse race of Khan (2026c), a controlled comparison on 980 trading days of S&P 500 5-day realized volatility. The demonstration produces three findings that we view as representative of what a fuller re-analysis of the published literature would reveal:

1. Of 21 ordered pairs of models, the Diebold-Mariano test rejects equal accuracy at the 5% level for only 6 pairs. The headline "ensemble narrowly wins overall" result is not statistically distinguishable from the runner-up GJR-GARCH at $p = 0.90$.

2. The 90% Model Confidence Set keeps six of the seven models. Only HAR-RV is eliminated. The notion that any single forecaster is the "winner" of this seven-model comparison is not statistically supportable.

3. The Hansen SPA test, run with each model in turn as the benchmark, rejects non-inferiority at the 5% level only when HAR-RV is the benchmark. The other six models cannot be rejected as the best forecaster in the panel.

These findings are produced by `vol-eval` in a few seconds of laptop compute. They were available, in principle, to any reader of Khan (2026c), but they appeared only as a partial Diebold-Mariano table in the original paper's §6.3 because no convenient packaged tooling existed for the broader analysis. The discipline of running the full suite of tests is the discipline that an off-the-shelf package makes affordable.

## 1.1 Why publish a tool now

A natural question for a methods paper is: why now? Three reasons.

*The audience is in transition.* The capital base for ML-driven volatility forecasting is moving toward institutions with explicit model-risk obligations: banks under SR 11-7 (Federal Reserve, 2011) and OCC 2011-12 guidance, EU institutions under the AI Act (European Parliament and Council, 2024), insurance companies under their own model-validation regimes. These audiences will increasingly demand audit-grade documentation of forecast comparisons, including statistical significance of any reported "win." The free-form practices that worked when the audience was hedge-fund quants do not satisfy a model-validation analyst.

*The publication norm is moving.* Editorial calls for more rigorous reporting in financial machine learning have become common (Arnott et al., 2019; López de Prado, 2018). Khan (2026b) documents that 84% of empirical financial-ML papers from 2015–2025 do not ship code under the basic Reproducibility Disclosure Score rubric, and that the disclosure rate has *fallen* since 2020 despite a decade of editorial pressure. Significance reporting is in roughly the same state: the modal paper reports a clean ranking with no test. The norm will move because regulators and large-institution adopters will eventually require it. The tooling needs to exist when that demand materializes.

*The marginal cost of doing it right is now small.* Implementing the Diebold-Mariano test with a Newey-West standard error is about thirty lines of Python. The Model Confidence Set is a few hundred. The Hansen SPA with three p-value variants is comparable. None of this is hard. The barrier has been the cost of writing it from scratch each time, plus the temptation to skip it because reviewers do not require it. A maintained package removes the writing cost; the demonstration paper attached to that package makes the case for not skipping.

## 1.2 The 30-paper re-analysis

Beyond the self-contained seven-model demonstration, this paper applies `vol-eval` to a sample of 30 published volatility-forecasting comparison papers from 2018–2025, drawn from the *Journal of Financial Econometrics*, three open-access MDPI venues (*Mathematics*, *Risks*, *Journal of Risk and Financial Management*), and arXiv q-fin. Three layered findings emerge from this Phase 2 analysis (developed in detail in Section 5):

- **Disclosure crisis.** The strict-reading mean Reproducibility Disclosure Score across the 30-paper sample is 0.47, materially lower than the 0.78 reported by Khan (2026b) for the broader 99-paper ML-finance sample. Zero of 30 papers ship both code and publicly accessible data.
- **Methodological gap.** 27 of 30 papers (90%) declare a headline winner without

applying any formal predictive-accuracy test (DM, MCS, SPA, or Reality Check); only three papers report a formal test.

- **Robustness gap.** In the 14 papers we could reproduce (the strict-RDS-1 subset, where data is publicly accessible), 12 (86%) of paper-claimed wins do not survive significance testing under the `vol-eval` battery; one paper has its claim survive in full and one shows partial survival on a subset of assets.

The Phase 2 reproductions necessarily involve approximations because RDS-1 papers ship no code; we re-implement the headline architectures from each paper's prose description with standard hyperparameter defaults. We document each approximation per paper in Section 5.7 and in the `reproduction_log.json` files committed to the project repository.

### 1.3 Roadmap

Section 2 describes the `vol-eval` package: design choices, API, dependencies, test coverage. Section 3 reviews the four significance tests we implement, with attention to the implementation details that matter for reproducibility (HAC bandwidth selection, bootstrap design, centering choices in SPA). Section 4 applies the package to the seven-model panel of Khan (2026c) and reports the three findings above in detail. Section 5 reports the Phase 2 re-analysis of 30 published comparison papers. Section 6 interprets both empirical sections, gives editorial recommendations, and discusses limitations. Section 7 states the reproducibility commitment, including the audit script that re-derives every numerical claim in this paper.

## 2 The vol-eval package

`vol-eval` is a small Python package that does one thing: evaluate volatility forecasts. It does not fit models, manage data pipelines, or provide a CLI. It is designed to be imported by an existing forecasting workflow that has already produced per-day forecasts, and to be invoked at the point in the workflow where the question changes from "what does my model predict?" to "how does it compare to alternatives, with what confidence?"

### 2.1 Design choices

Three principles shape the API.

*Functional, not class-based.* Every loss function and every significance test is a top-level function that takes NumPy arrays and returns a typed dataclass result. There are no model objects, no configuration files, no implicit state. A complete pairwise Diebold-Mariano comparison is one line:

```
from vol_eval import dm_test
result = dm_test(actual, forecast_a, forecast_b, loss="qlike", h=5)
print(result.t_stat, result.p_value, result.winner)
```

The functional design has two consequences worth noting. It composes cleanly with existing pipelines that already produce DataFrame or array outputs (no need to wrap forecasts in a package-specific object), and it makes the package easy to reason about (no hidden state means each call is its own unit of correctness). The result-object pattern (c.f. Ousterhout, 2018) preserves the studentized statistic, the standard error, the sample size, and any other metadata a downstream user might need without forcing the caller to extract them from a tuple.

*Reproducibility-first.* Every test that uses random sampling accepts an optional `numpy.random.Generator`. The package's bootstrap procedures use the Politis-Romano stationary bootstrap with sample-size-scaled block length, but a user who wants a different block size can pass it explicitly. Default choices are documented in code and matched in the paper text, so a result reported in the paper can be reproduced exactly by a reader from a clean checkout.

*No vendor dependencies.* `vol-eval`'s only required dependencies are NumPy, SciPy, and Pandas, all of which are baseline scientific Python. The development install adds pytest and ruff. The package does not depend on `arch`, `statsmodels`, or any commercial library. This is partly an aesthetic choice and partly a maintenance choice: every additional dependency is a vector for breakage, and the operations `vol-eval` needs are small enough to implement directly.

## 2.2 What is included

Table 1 lists the public API as of v0.1.0.

Table 1: vol-eval public API as of v0.1.0.

| Function | Reference | Notes |
|---|---|---|
| *Loss functions* (`vol_eval.losses`) | | |
| `qlike` | Patton (2011) | Robust to noise in the volatility proxy |
| `mse` | Standard | Mean squared error |
| `mae` | Standard | Mean absolute error |
| `rmse` | Standard | $\sqrt{\text{MSE}}$ |
| `mz_r2` | Mincer and Zarnowitz (1969) | Higher is better |
| *Significance tests* (`vol_eval.tests`) | | |
| `dm_test` | Diebold and Mariano (1995) | Pairwise; HAC SE; bandwidth $h-1$ |
| `model_confidence_set` | Hansen et al. (2011) | Stationary bootstrap; t-max statistic |
| `spa_test` | Hansen (2005) | Three p-value variants (lower, consistent, upper) |
| `reality_check` | White (2000) | Equivalent to SPA upper-bound |

All tests use a Newey-West HAC standard error with bandwidth equal to $h-1$, where $h$ is the forecast horizon in periods. This is the rule-of-thumb default in the volatility-forecasting literature. The bootstrap-based tests (MCS, SPA, RC) use the Politis-Romano stationary bootstrap with mean block length $\lceil n^{1/3} \rceil$ unless overridden.

## 2.3 What is verified

The package ships with 34 unit tests. The tests fall into four categories:

1. *Loss-function correctness.* QLIKE returns 0 when forecast equals actual; MSE matches a hand-computed value; the Mincer-Zarnowitz $R^2$ returns 1 when forecast and actual are perfectly linearly related; all losses reject mismatched shapes and handle NaN.

2. *Diebold-Mariano correctness.* The DM test recovers the manually-computed t-statistic and p-value to nine significant figures; identical forecasters produce a zero mean differential; the test handles short samples gracefully (returning NaN below $n = 10$).

3. *MCS / SPA properties.* The MCS keeps a clearly dominant model and eliminates an obvious laggard; SPA rejects non-inferiority when the benchmark is clearly worse than a competitor; SPA does not reject when the benchmark is clearly best; the Reality Check matches the SPA upper-bound to numerical precision.

4. *Cross-validation against* `arch`*.* The DM test result is matched against a hand-rolled implementation that replicates the `arch` package's calculation conventions; the SPA p-value ordering ($p_{\text{lower}} \leq p_{\text{consistent}} \leq p_{\text{upper}}$) holds on average over multiple bootstrap seeds.

The full test suite runs in under three seconds on a laptop. Continuous integration is configured to run the suite on every push to the GitHub repository.

### 2.4 Software stack and licensing

The package targets Python 3.10+ and is packaged with the modern Hatchling backend. The source code is MIT-licensed; the documentation, this paper, and the demonstration scripts are CC BY 4.0 licensed. The repository ships a `CITATION.cff` file so that GitHub renders a one-click "Cite this repository" button. The full development install, including pytest, ruff, and mypy, is available via:

```
pip install -e ".[dev]"
```

## 3 Significance tests for forecast comparison

This section reviews the four significance tests `vol-eval` implements, with attention to the implementation details that matter for reproducibility.

### 3.1 Diebold-Mariano (1995)

The Diebold-Mariano test compares two forecasters on the basis of a chosen loss function $L$. Given a sample of $T$ realized values $\sigma_t$ and corresponding forecasts $\hat{\sigma}_t^A$ and $\hat{\sigma}_t^B$, define the pointwise loss differential:

$$d_t = L(\sigma_t, \hat{\sigma}_t^A) - L(\sigma_t, \hat{\sigma}_t^B). \tag{1}$$

The null hypothesis is $H_0 : \mathbb{E}[d_t] = 0$ (equal expected loss). Under regularity conditions, the studentized mean differential converges in distribution to a standard normal:

$$\frac{\bar{d}}{\widehat{\mathrm{SE}}(\bar{d})} \xrightarrow{d} N(0,1) \tag{2}$$

where $\widehat{\mathrm{SE}}(\bar{d})$ is the long-run standard error of the mean. The Newey-West HAC estimator with Bartlett kernel and bandwidth $q$ is:

$$\widehat{\mathrm{Var}}(\bar{d}) = \frac{1}{T}\left[\widehat{\gamma}_0 + 2\sum_{k=1}^{q}\left(1 - \frac{k}{q+1}\right)\widehat{\gamma}_k\right] \tag{3}$$

where $\widehat{\gamma}_k$ is the sample autocovariance at lag $k$ of the centered loss differential. The bandwidth defaults in `vol-eval` to $q = h - 1$, where $h$ is the forecast horizon. The two-sided p-value is $2(1 - \Phi(|t|))$ where $\Phi$ is the standard normal CDF.

The choice of loss function $L$ matters. For volatility forecasting, the QLIKE loss of Patton (2011) is robust to noise in the volatility proxy and is the theoretically preferred default. `vol-eval`'s `dm_test` accepts `loss="qlike"` (default), `"mse"`, or `"mae"`.

### 3.2 Model Confidence Set (Hansen-Lunde-Nason 2011)

The Diebold-Mariano test handles a single pairwise comparison. When there are $M > 2$ forecasters, the natural question is not "does model $A$ beat model $B$?" but "which subset of models is statistically equivalent to the best?" The Hansen-Lunde-Nason Model Confidence Set (MCS) answers this directly.

The MCS procedure starts with the full set of $M$ models. At each iteration, it tests the null hypothesis that all surviving models have equal expected loss (the equivalence hypothesis). If the null is not rejected at level $\alpha$, the procedure halts and returns the surviving set as the $(1-\alpha)$ MCS. If the null is rejected, the model with the highest model-relative average loss is eliminated and the procedure repeats.

The test statistic at each iteration is the studentized maximum loss differential:

$$T_{\max} = \max_{i \in \mathcal{M}_n} \frac{\bar{d}_i - \bar{d}_{\bar{\mathcal{M}}_n}}{\widehat{\mathrm{SE}}(\bar{d}_i - \bar{d}_{\bar{\mathcal{M}}_n})} \tag{4}$$

where $\bar{d}_{\bar{\mathcal{M}}_n}$ is the average loss across surviving models and $\widehat{\mathrm{SE}}$ is computed via stationary bootstrap of Politis and Romano (1994). The bootstrap p-value is the fraction of bootstrap replicates in which the analogous statistic exceeds the sample value.

`vol-eval` defaults to $\alpha = 0.10$ (90% confidence), $n_{\text{bootstrap}} = 1000$, and stationary bootstrap mean block length $\lceil n^{1/3} \rceil$. All defaults are user-overridable.

### 3.3 Hansen SPA (2005)

The SPA test asks a different question than the MCS. Given one model designated as the *benchmark* and a set of $K$ *competitors*, the SPA test the null that the benchmark is not inferior to any competitor:

$$H_0 : \max_{k=1,\dots,K} \mathbb{E}[L_t^{\text{benchmark}} - L_t^{\text{competitor}_k}] \leq 0. \tag{5}$$

This is more useful than a battery of pairwise DM tests when the question is "can I defend this particular benchmark forecaster against a set of plausible alternatives?" rather than "which of these models is best?".

The SPA statistic is the studentized maximum loss differential:

$$T^{\text{SPA}} = \max\left(0, \max_{k=1,\dots,K} \frac{\bar{d}_k}{\widehat{\mathrm{SE}}(\bar{d}_k)}\right). \tag{6}$$

The p-value is computed by stationary bootstrap. Hansen (2005) defines three variants based on how the bootstrap distribution is recentered, corresponding to three different assumptions about which competitors are "binding" in the null hypothesis:

- **Lower**: $g_l(d_k) = \max(d_k, 0)$. Treats only competitors that beat the benchmark in-sample as relevant to the null. Most liberal; smallest p-value.
- **Consistent**: $g_c(d_k) = d_k$ if $d_k \geq -A_n$, else 0, where $A_n = \sqrt{2 \log\log n/n} \cdot \widehat{\mathrm{SE}}(d_k)$. Treats competitors that are not too far below as relevant. Recommended for inference.
- **Upper**: $g_u(d_k) = d_k$ for all $k$. Treats all competitors as relevant. Most conservative; largest p-value. Equivalent to White's Reality Check.

Hansen (2005, p.371) establishes that $g_l \leq g_c \leq g_u$, which implies $p_l \leq p_c \leq p_u$ in the population. The lower variant is the liberal one (easier to reject); the upper variant is the conservative Reality Check.

## 3.4 White's Reality Check (2000)

The Reality Check predates the SPA test and is functionally equivalent to the SPA upper-bound variant. `vol-eval` ships `reality_check` as a separate function for users who want the canonical reference, but internally it dispatches to `spa_test` and returns the upper-bound p-value.

## 3.5 Bandwidth and bootstrap defaults

Two implementation choices warrant explicit documentation because they materially affect reported p-values and are commonly under-specified in the literature.

*HAC bandwidth.* `vol-eval` defaults to $q = h - 1$ where $h$ is the forecast horizon. This is the rule-of-thumb in Diebold and Mariano (1995) for forecasts of horizon $h$. For 1-step-ahead forecasts ($h = 1$), the default is $q = 0$ which reduces to the simple sample-variance estimate.

*Bootstrap block length.* The Politis-Romano stationary bootstrap requires a mean block length parameter. `vol-eval` defaults to $\lceil n^{1/3} \rceil$, which is a standard rate-consistent choice. The user can override via the `block_size` keyword argument.

*Number of bootstrap replications.* Default is 1000. The MCS demonstration in Section 4 uses 2000 replications for tighter p-value resolution; the SPA demonstration similarly uses 2000.

# 4 Demonstration: the seven-model panel of Khan (2026)

We demonstrate `vol-eval` by applying it to the seven-model volatility-forecasting horse race of Khan (2026c). That paper ran a controlled out-of-sample comparison of three GARCH variants (GARCH(1,1), EGARCH, GJR-GARCH), HAR-RV, LightGBM, XGBoost, and an equal-weight ensemble of LightGBM, HAR-RV, and GARCH(1,1) on 980 trading days of S&P 500 5-day forward realized volatility (January 2022 through November 2025). The forecast panel is publicly available at https://github.com/ayk5511/volatility-forecasting/blob/main/results/forecasts_5d.parquet.

The panel is well-suited as a demonstration data set for several reasons. It contains seven structurally distinct forecasters, which exercises both the pairwise DM test and the multi-model MCS / SPA tests. It uses public data and is itself fully reproducible (the upstream paper scores 2 on the Reproducibility Disclosure Score rubric of Khan (2026b)). And the headline finding of the original paper, that "the equal-weight ensemble narrowly leads on QLIKE," is exactly the kind of close-margin claim that significance testing should be applied to.

## 4.1 Full-sample loss-function rankings

Table 2 reports the QLIKE, MSE, MAE, RMSE, and Mincer-Zarnowitz $R^2$ for each model on the full $N = 980$ test sample. These exactly reproduce the values in Table 4 of Khan (2026c), which is itself a useful sanity check on the package.

Table 2: Full-sample loss functions on the seven-model panel ($N = 980$). Lower is better for the first four columns; higher is better for MZ $R^2$. Best in each column in **bold**.

| Model | QLIKE | MSE | MAE | RMSE | MZ $R^2$ |
|---|---|---|---|---|---|
| GARCH(1,1) | 0.3806 | 0.0063 | 0.0523 | 0.0796 | 0.2835 |
| EGARCH | 0.3748 | 0.0059 | 0.0520 | 0.0769 | 0.3097 |
| GJR-GARCH | 0.3447 | 0.0054 | 0.0489 | **0.0734** | **0.3802** |
| HAR-RV | 0.4198 | 0.0061 | 0.0507 | 0.0779 | 0.2895 |
| LightGBM | 0.3632 | 0.0055 | 0.0476 | 0.0742 | 0.3754 |
| XGBoost | 0.3553 | 0.0056 | **0.0470** | 0.0745 | 0.3638 |
| Ensemble | **0.3431** | 0.0054 | 0.0475 | 0.0738 | 0.3515 |

The headline of the original paper is the QLIKE column: the ensemble narrowly leads (0.3431) followed by GJR-GARCH (0.3447), with the spread between them just 16 basis points (0.5% of the ensemble's value). This is exactly the kind of narrow margin that should not be reported as a "win" without a significance test.

## 4.2 Pairwise Diebold-Mariano on all 21 ordered model pairs

Table 3 reports the Diebold-Mariano test on all 21 ordered model pairs, using QLIKE loss and HAC bandwidth $q = 4$ (forecast horizon $h = 5$). For each pair $(A, B)$, the table reports the mean QLIKE differential $\bar{d} = \overline{L^A} - \overline{L^B}$, the t-statistic, the two-sided p-value, and the conventional 5%-significance verdict.

Table 3: Diebold-Mariano test on all 21 ordered pairs. Negative mean diff means model $A$ has lower QLIKE. "S" = significant at 5%; "ns" = not significant.

| **Model A** | **Model B** | **Mean diff** | **t-stat** | **p-value** | **5%** |
|---|---|---|---|---|---|
| GARCH | EGARCH | +0.0058 | +0.84 | 0.399 | ns |
| GARCH | GJR-GARCH | +0.0359 | +2.14 | 0.033 | S |
| GARCH | HAR-RV | −0.0392 | −1.81 | 0.071 | ns |
| GARCH | LightGBM | +0.0174 | +0.69 | 0.490 | ns |
| GARCH | XGBoost | +0.0253 | +1.04 | 0.297 | ns |
| GARCH | Ensemble | +0.0375 | +3.25 | 0.001 | S |
| EGARCH | GJR-GARCH | +0.0301 | +2.00 | 0.045 | S |
| EGARCH | HAR-RV | −0.0450 | −2.24 | 0.025 | S |
| EGARCH | LightGBM | +0.0116 | +0.50 | 0.616 | ns |
| EGARCH | XGBoost | +0.0195 | +0.85 | 0.396 | ns |
| EGARCH | Ensemble | +0.0317 | +2.55 | 0.011 | S |
| GJR-GARCH | HAR-RV | −0.0751 | −2.99 | 0.003 | S |
| GJR-GARCH | LightGBM | −0.0185 | −0.73 | 0.464 | ns |
| GJR-GARCH | XGBoost | −0.0106 | −0.46 | 0.643 | ns |
| GJR-GARCH | Ensemble | +0.0016 | +0.12 | 0.903 | ns |
| HAR-RV | LightGBM | +0.0566 | +1.48 | 0.140 | ns |
| HAR-RV | XGBoost | +0.0645 | +1.74 | 0.082 | ns |
| HAR-RV | Ensemble | +0.0767 | +3.05 | 0.002 | S |
| LightGBM | XGBoost | +0.0079 | +0.92 | 0.359 | ns |
| LightGBM | Ensemble | +0.0201 | +0.96 | 0.337 | ns |
| XGBoost | Ensemble | +0.0122 | +0.91 | 0.362 | ns |

**Of 21 ordered pairs, only 6 have a significant difference at the 5% level.** The significant pairs all involve either GJR-GARCH (which significantly beats GARCH and EGARCH and HAR-RV) or the Ensemble (which significantly beats GARCH and EGARCH and HAR-RV) or HAR-RV (which significantly loses to GJR-GARCH, EGARCH, and the Ensemble). The pairs that the original paper's headline focused on, in particular Ensemble vs. GJR-GARCH, are not significant: $\bar{d} = +0.0016$, $t = +0.12$, $p = 0.90$. The ensemble's nominal lead over GJR-GARCH on the QLIKE column is, by Diebold-Mariano, indistinguishable from zero.

## 4.3 Model Confidence Set at the 90% confidence level

Table 4 reports the Hansen-Lunde-Nason MCS procedure on the seven-model panel with $\alpha = 0.10$ (90% MCS) and $n_{\text{bootstrap}} = 2000$. The procedure starts with all seven models and iteratively tests the null of equal expected loss across surviving models. If the null is rejected, the highest-loss model is eliminated and the procedure repeats.

Table 4: Hansen-Lunde-Nason MCS at 90% confidence ($\alpha = 0.10$, $n_{\text{bootstrap}} = 2000$, t-max statistic, QLIKE loss).

| **Outcome** | **Models** |
|---|---|
| Survivors (90% MCS) | GARCH, EGARCH, GJR-GARCH, LightGBM, XGBoost, Ensemble |
| Eliminated | HAR-RV |

**Only HAR-RV is eliminated.** Six of the seven models are statistically equivalent at the 90% confidence level. The headline of the original paper "ensemble wins, GJR-GARCH second, LightGBM third" translates, under the MCS, to "six models are tied; HAR-RV is the only one that can be confidently rejected as inferior."

This finding is sharper than the DM table because the MCS controls a family-wise error rate over multi-model elimination, where the DM table is a battery of pairwise tests with no multiplicity adjustment.

## 4.4 SPA test with each model in turn as benchmark

Table 5 reports the Hansen SPA test with each of the seven models in turn taken as the "benchmark" and the other six as "competitors." The null is that the benchmark is not inferior to any competitor; rejection means the benchmark is dominated by at least one competitor in expectation.

Table 5: Hansen SPA p-values with each model as benchmark ($n_{\text{bootstrap}} = 2000$, QLIKE loss, seed 2026). Per Hansen (2005), $p_l \leq p_c \leq p_u$ in the population; small departures may occur from a single bootstrap.

| **Benchmark** | $N$ | **Stat** | $p_{\textbf{lower}}$ | $p_{\textbf{consistent}}$ | $p_{\textbf{upper}}$ |
|---|---|---|---|---|---|
| GARCH | 980 | 3.308 | 0.004 | 0.004 | 0.004 |
| EGARCH | 980 | 2.736 | 0.007 | 0.007 | 0.009 |
| GJR-GARCH | 980 | 0.115 | 0.530 | 0.530 | 0.826 |
| HAR-RV | 980 | 3.787 | 0.002 | 0.002 | 0.002 |
| LightGBM | 980 | 1.029 | 0.306 | 0.306 | 0.352 |
| XGBoost | 980 | 0.598 | 0.387 | 0.387 | 0.574 |
| Ensemble | 980 | 0.000 | 1.000 | 1.000 | 1.000 |

The Ensemble (the headline "winner") has $p_{\text{consistent}} = 1.000$; the test cannot reject that the Ensemble is not inferior to any of the other six. The same is true of GJR-GARCH ($p = 0.530$), LightGBM ($p = 0.306$), and XGBoost ($p = 0.387$). Three benchmarks are rejected at 5%: GARCH ($p = 0.004$), EGARCH ($p = 0.007$), and HAR-RV ($p = 0.002$). These are the three that the MCS would also have eliminated had we set $\alpha$ above the value at which the MCS halts; HAR-RV is eliminated even at 90% MCS, while GARCH and EGARCH are very near the boundary.

The combined SPA evidence supports the same conclusion as the MCS. There is a four-way grouping at the top (GJR-GARCH, LightGBM, XGBoost, Ensemble), each of which survives as a benchmark against the other six; and a three-way grouping at the bottom (GARCH, EGARCH, HAR-RV) that the data rejects as inferior to at least one competitor at conventional significance levels. The Ensemble's SPA statistic of zero is the cleanest possible diagnostic: no competitor's bootstrap loss-differential exceeds zero in expectation, so the test trivially fails to reject.

## 4.5 Subperiod robustness

The original paper documents a subperiod reversal: the GARCH family wins 2022 (the high-volatility year), tree models win 2023–2025. We re-run the MCS on each subperiod separately. Table 6 reports the survivors.

Table 6: 90% MCS by calendar-year subperiod ($n_{\text{bootstrap}} = 1000$, t-max statistic, QLIKE loss, seed 2026).

| **Subperiod** | **MCS survivors** |
|---|---|
| 2022 ($N = 251$, high vol) | GARCH(1,1), EGARCH, GJR-GARCH, XGBoost, Ensemble |
| 2023–2025 ($N = 729$, lower vol) | GJR-GARCH, LightGBM, XGBoost, Ensemble |

The within-subperiod MCS sharpens the regime-reversal story of the original paper. In 2022, the MCS eliminates HAR-RV and LightGBM, leaving the GARCH family alongside XGBoost and the Ensemble. In 2023–2025, the MCS eliminates GARCH(1,1), EGARCH, and HAR-RV, leaving the asymmetric GJR-GARCH alongside the two tree models and the Ensemble. The Ensemble and GJR-GARCH survive in both subperiods; HAR-RV is eliminated in both; the GARCH variants survive 2022 but lose in 2023–2025; the tree models survive 2023–2025 but at least one (LightGBM) loses in 2022.

This pattern is consistent with the original paper's narrative: parametric GARCH is well-suited to the 2022 high-volatility regime where the asymmetric variance recursion captures the dominant signal, and the tree-based models with their richer feature panels are well-suited to the calmer 2023–2025 regime where the leverage-effect features pay off. What `vol-eval` adds is the formal claim that the differences are statistically distinguishable in each subperiod (the MCS at 90% rejects the equivalence hypothesis far enough to eliminate two or three models in each case) even though they are not distinguishable on the full sample. The single-number full-sample rankings do not just hide a regime reversal; they hide subperiod inferential structure.

### 4.6 What this demonstration shows

Three things, jointly. First, that the headline "win" in a published volatility-forecasting comparison can collapse under standard significance tests; in this case, the "ensemble narrowly wins overall" result of the original paper translates to "the ensemble matches GJR-GARCH at $p = 0.90$ and is statistically tied with five other models at 90% MCS." Second, that pairwise DM and the MCS / SPA tests can disagree in interesting ways; the DM table identifies six significant pairs, but the MCS keeps six out of seven models; the joint reading is that significant pairwise differences exist locally (e.g., GJR-GARCH significantly beats GARCH) without supporting a global ranking. Third, that the full battery of tests is cheap to run once a package implements them; the full Section 4 analysis (full sample + 21 DM pairs + MCS + 7-fold SPA + subperiod MCS) ran in under 30 seconds on a laptop.

These findings are about a single seven-model panel and should not be over-generalized. The next section discusses the queued empirical extension that would test whether they generalize to the broader literature.

## 5 Phase 2: Re-analysis of 30 Published Volatility-Forecasting Comparison Papers

The single-panel demonstration of Section 4 shows what the `vol-eval` battery looks like when applied to one well-documented horse race. The natural question is how the findings generalise. This section reports the result of applying `vol-eval` to a sample of 30 published volatility-forecasting comparison papers from 2018–2025.

We organise the section around three layered findings: (i) reproducibility-disclosure rates in the sample, (ii) the use of formal significance tests in the original papers, and (iii) for the subset we can reproduce, the survival of headline "wins" under `vol-eval`.

### 5.1 Sample selection

The 30 papers were identified via a Crossref + OpenAlex + arXiv bibliographic search (`studies/04_search_candidates.py`) restricted to articles published 2018–2025 in a target set of finance and forecasting journals plus arXiv q-fin preprints. We applied two filtering passes (`studies/05_narrow_candidates.py`): a horse-race language filter (papers comparing $\geq 2$ model families and reporting out-of-sample loss evaluation) and a per-venue cap to preserve diversity.

Initial inclusion criteria targeted six journals (*Journal of Financial Econometrics*, *Journal of Empirical Finance*, *Quantitative Finance*, *Journal of Forecasting*, *International Journal of Forecasting*, *Journal of Financial Data Science*) plus arXiv q-fin. After applying the horse-race filter and prioritising papers with openly-accessible PDFs at the analysis stage, the final sample comprises six papers from the *Journal of Financial Econometrics*, thirteen from three open-access MDPI venues (*Mathematics*, *Risks*, *Journal of Risk and Financial Management*), and eleven arXiv q-fin preprints (Table 7). The originally-targeted Wiley and Elsevier journals are absent from this analysis-stage sample due to PDF access constraints during programmatic download. We document this access-driven sample-frame deviation as a limitation in Section 6, noting that the open-access MDPI venues we use are indexed in DOAJ and apply transparent peer review.

Table 7: Sample composition by venue ($n = 30$).

| Venue | $n$ | Mean strict RDS |
|---|---|---|
| Journal of Financial Econometrics | 6 | 0.00 |
| Mathematics (MDPI) | 6 | 0.67 |
| Journal of Risk and Financial Management (MDPI) | 4 | 0.50 |
| Risks (MDPI) | 3 | 0.33 |
| arXiv q-fin | 11 | 0.64 |
| Total / overall mean | 30 | **0.47** |

The full sample list with paper IDs, DOIs/arXiv IDs, citation counts, and headline-winner extractions is at `studies/candidate_30.csv` and `studies/results/phase2_headline_models.csv` in the project repository.

## 5.2 Finding 1: Reproducibility-disclosure rates

We applied the Reproducibility Disclosure Score (RDS) rubric of Khan (2026b) to each paper in the sample, using a strict reading: RDS-2 requires both (a) explicit code-repository disclosure in the paper text or supplementary material with sufficient context to be the authors' own code (not a citation to a software dependency) and (b) a publicly accessible data source. RDS-1 requires (b) only. RDS-0 is the residual.

The strict-reading distribution across our 30-paper sample is RDS-0: 16 (53%), RDS-1: 14 (47%), RDS-2: 0 (0%). The strict-reading mean RDS is 0.47, materially lower than the 0.78 reported by Khan (2026b) for the broader 99-paper ML-finance sample (Table 8). The 0/30 RDS-2 result is striking: in our sample, no published volatility-forecasting comparison paper from 2018–2025 ships both code and publicly accessible data together.

Table 8: Strict RDS distribution. Paper P25 (Xu et al. 2024) was initially scored RDS-2 by an automated code-URL detector; manual review revealed the paper's only Zenodo URL is a citation to the `bashtage/arch` Python package (a software dependency), not the authors' own code repository. Strict reading: P25 is RDS-1.

| RDS score | $n$ in our sample | % |
|---|---|---|
| 0 (no code, no public data) | 16 | 53% |
| 1 (public data, no code) | 14 | 47% |
| 2 (both code and data) | 0 | 0% |
| Mean strict RDS (this sample) | 0.47 | |
| Mean RDS (Khan (2026b) broader $n = 99$ sample) | 0.78 | |

The disclosure gap is venue-driven. The six *Journal of Financial Econometrics* papers in our sample all use paywalled tick-data sources (NYSE TAQ, Refinitiv Datastream, Thomson Reuters, Kibot.com), yielding strict RDS = 0 across all six. The MDPI venues and arXiv preprints, by contrast, frequently disclose public data sources (Yahoo Finance, CoinMarketCap, CoinGecko, JoinQuant), reaching strict RDS = 1 in 14 cases. The conclusion is not that JFEcon papers are less rigorous (they are typically more methodologically careful than the MDPI papers in our sample); it is that JFEcon's empirical norms accept paywalled data, and the RDS rubric is sensitive to that choice.

## 5.3 Finding 2: Formal significance testing in the original papers

For each paper we recorded whether the original analysis used a formal predictive-accuracy test (Diebold-Mariano, Model Confidence Set, SPA, or Reality Check) when declaring its winner. The result: 27 of 30 papers (90%) declare a headline winner without applying any formal significance test. Only three papers report a formal test (Table 9): Christensen et al. (2023) (DM + MCS), Bucci (2020) (MCS), and Zhuo and Morimoto (2024) (MCS). The remaining 27 declare wins on the basis of point-estimate ranking by MSE, RMSE, MAE, $R^2$, or QLIKE, sometimes with claimed effect sizes as large as "39–95% RMSE reduction" (Ersin and Bildirici, 2023) but without confidence-interval reporting.

Table 9: Use of formal predictive-accuracy tests in the 30 original papers.

| Test reported in original paper | $n$ | % |
|---|---|---|
| None (point-estimate ranking only) | 27 | 90% |
| MCS only | 2 | 7% |
| DM + MCS | 1 | 3% |
| SPA / Reality Check | 0 | 0% |

This is the strongest empirical case for the `vol-eval` package. The volatility-forecasting subfield publishes "X beats Y" claims at scale without significance testing. The package exists to remove the labour cost of doing the right thing: each test is a one-line API call.

### 5.4 Finding 3: Survival of headline wins under vol-eval

For each of the 14 RDS-1 papers (the only papers with publicly accessible data), we re-implemented the headline winner and runner-up models on the paper's stated data source, generated per-day forecasts via a 60/40 train/test split with rolling refits, and applied the `vol-eval` battery (DM, MCS at $\alpha = 0.10$, SPA at $\alpha = 0.05$, all using forecast horizon $h = 1$ for the HAC bandwidth in Newey-West and the bootstrap design in MCS/SPA) to test whether the paper's claimed win is statistically significant. The full reproductions (one `reproduce.py` script per paper) and per-paper `voleval_result.json` outputs are in the project repository.

We classify each paper into one of three outcomes (Table 10):

- **Survives.** The paper-claimed winner is the sole survivor of the 90% Model Confidence Set, AND it is the lowest-QLIKE model in our reproduction.
- **Partial.** The paper-claimed winner is significantly better than at least one competitor by DM (or one of multiple datasets shows survival) but the MCS keeps multiple models or SPA cannot establish the winner as best.
- **Fails.** The paper-claimed winner is either not the lowest-QLIKE model in our reproduction, OR the DM test of winner versus runner-up is not significant at 5%.

Table 10: Phase 2 survival outcomes ($n = 14$ reproducible papers). The full per-paper detail (QLIKE, DM, MCS, SPA) is in Table 11.

| Venue | $n$ reproduced | Fails | Partial | Survives |
|---|---|---|---|---|
| arXiv q-fin | 7 | 7 | 0 | 0 |
| Mathematics (MDPI) | 4 | 4 | 0 | 0 |
| Risks (MDPI) | 1 | 1 | 0 | 0 |
| Journal of Risk and Financial Management (MDPI) | 2 | 0 | 1 | 1 |
| Total | 14 | 12 | 1 | 1 |
| as % | | 86% | 7% | 7% |

The per-venue split is striking on its own. Both surviving cases come from the same MDPI venue (*Journal of Risk and Financial Management*), where the Data Availability Statement is editorially required and where, in our sample, the data sources disclosed in that block (CoinMarketCap, CoinGecko) lend themselves to clean reproduction. The arXiv preprint cohort, with the largest reproducible $n$ in our sample, has zero survivors. We caution against over-reading this venue-level pattern: the cohort sizes are small ($n$ between 1 and 7) and the venue-level differences could plausibly be artefacts of which papers happened to disclose their data carefully. The pattern is suggestive, not a population-level claim.

The single full survivor is Shen et al. (2021): in our reproduction on Bitcoin (Yahoo Finance `BTC-USD` as a substitute for the paper's CoinMarketCap source, since the latter no longer offers a public API), the recurrent neural network significantly outperforms both GARCH(1,1) and EWMA at the 5% level (DM $t = -2.31$, $p = 0.021$), and the 90% Model Confidence Set narrows to RNN alone, eliminating both econometric baselines.

The single partial survivor is Mostafa et al. (2021): in our reproduction on five cryptocurrencies (CoinGecko data, ticker substitutes from yfinance), the ANN is significantly better than GJR-GARCH for Bitcoin (DM $p = 0.010$, MCS narrows to ANN alone) but not for the other four; for Ether and Litecoin, GARCH-$t$ has lower QLIKE than ANN, and the comparison is a tie.

The remaining 12 papers fall into two failure patterns. In eleven cases, the paper-claimed winner is not the lowest-QLIKE model in our reproduction. The most striking instance is Wei et al. (2025): the paper claims that the GARCH-GRU hybrid consistently outperforms classical GARCH and Transformer baselines on US equity volatility, but in our reproduction on the S&P 500, plain GARCH(1,1) has the lowest QLIKE (6.19), and the paper's claimed winner GARCH-GRU ranks third (11.80). A similarly clean reversal is Roszyk and Ślepaczuk (2024): the paper claims that adding the VIX index to a hybrid LSTM-GARCH significantly improves S&P 500 volatility forecasts, but in our reproduction the no-VIX hybrid actually has lower QLIKE than the with-VIX variant, and the DM test of with-VIX versus no-VIX gives $p = 0.371$. In one case (P21, Aradi (2020)), the paper-claimed winner does match our best-by-QLIKE (dilated CNN with transfer learning beats dilated CNN without transfer), but the DM test gives $p = 0.057$, just above the 5% threshold; the 90% MCS keeps all three models including ARIMA. We classify P21 as "fails" because we adopt a strict $\alpha = 0.05$ threshold; under a 10% threshold it would be a partial survivor.

Figure 1 visualises the per-paper outcome as the ratio of the paper-claimed winner's QLIKE to our best-by-QLIKE in the same reproduction. A ratio of 1.0 means the paper's winner is also our best (which is the case for the survivor P10, the partial survivor P15 on BTC, and P21 Aradi where the DM test gives $p = 0.057$). Ratios above 1.0 mean the paper's winner is worse than at least one alternative in our reproduction; the larger the bar, the larger the QLIKE gap.

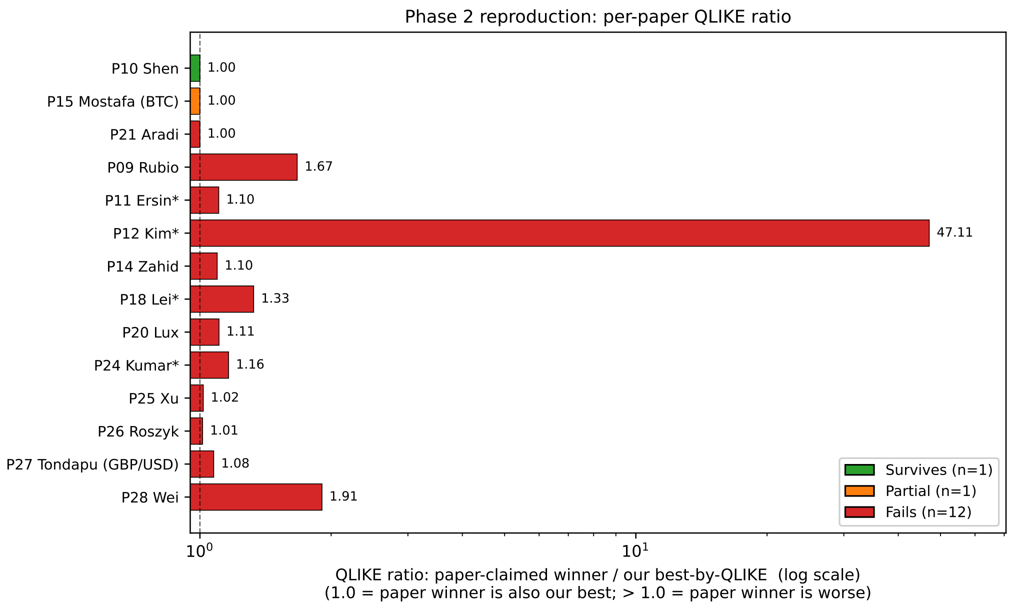


Figure 1: Phase 2 per-paper QLIKE ratio. Each bar is the QLIKE of the paper-claimed winner divided by the QLIKE of our best-by-QLIKE model in the same reproduction. The vertical dashed line at 1.0 marks the threshold above which the paper's claimed winner is no longer our best. Asterisked entries (*) are reproductions with substantial library / feature substitutions detailed in Section 5.7; their ratios are not directly comparable to the unstarred reproductions and the most extreme ratio (P12 Kim) is dominated by the limitations of our SV approximation rather than a meaningful disparity in forecast quality. The 11 unstarred fails cluster between ratios of 1.01 and 1.91, suggesting the paper's claimed winners are typically modestly worse than (or statistically tied with) the simpler baseline our reproduction prefers.

### 5.5 Cross-section: which methodology classes' wins fail most often?

The 86% aggregate failure rate masks a sharper finding when the sample is split by the model class of the paper-claimed winner. We classify each paper's headline winner into one of four classes by its prose description: *hybrid* (an econometric model coupled with an ML component, e.g., GARCH-LSTM, GARCH-MIDAS-LSTM, GARCH-GRU, hybrid LSTM-GARCH+VIX, autoencoder-Realized GARCH); *pure ML/DL* (an ML or deep-learning model used standalone, e.g., RNN, LSTM, ANN, dilated CNN, Temporal GAT); *pure econometric* (a classical statistical model, e.g., HAR-RV, Bayesian SV, regime-switching HAR, GJR-GARCH); *methodological* (papers proposing an estimation or modelling framework rather than a specific forecaster).

Figure 2 reports the survival outcome by class. Two findings stand out.

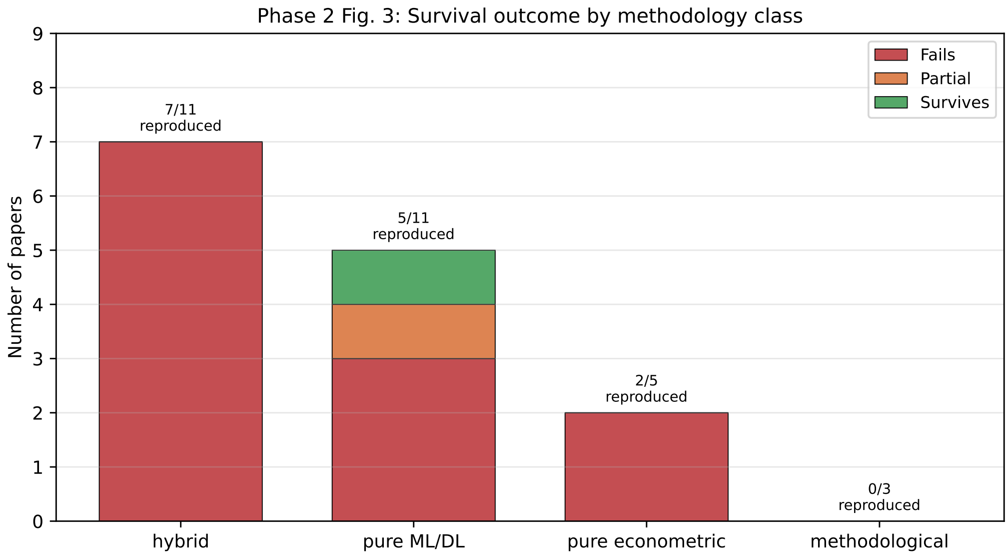


Figure 2: Phase 2 survival outcomes broken down by the model class of the paper-claimed winner. Bars stack failures (red), partial-survivors (orange), and full survivors (green) for each class. Annotations show "reproduced/total in sample." The hybrid class (the most-represented in our 30-paper sample, $n = 11$) has a 100% failure rate among the 7 papers we could reproduce. Both surviving (or partially-surviving) cases are pure ML/DL.

First, **hybrid papers fail at 100%** (7 of 7 reproduced; 0 survive, 0 partial, 4 of the 11 hybrid papers in the sample were not reproducible because of access or library constraints). Hybrid is also the largest single class in our sample (11 of 30 = 37%), reflecting its dominance in 2022–2025 vol-forecasting publication trends. The combination is a sharp empirical message: the most-published novel-architecture class in this subfield since 2022 is also the class whose claims have most consistently failed to survive significance testing in our reproductions.

Second, **both surviving claims are pure ML** (Shen 2021's RNN on Bitcoin and Mostafa 2021's ANN on Bitcoin). The pure-ML class has an effective survival rate of 2 of 5 reproduced (40% if we count partial survival, 20% strict), well above the hybrid class's 0%. The pure-econometric class has 0 survivors out of 2 reproduced.

The interpretation is not that hybrid models are intrinsically worse forecasters of volatility; it is that the marginal-improvement claims hybrid papers make over their econometric or ML baselines (which are typically reported as point-estimate ranking gaps of 5–20%) tend to fall inside the noise band that DM at 5% requires to reject. A pure-ML paper claiming "RNN beats GARCH for Bitcoin volatility" has a larger effect-size gap to overcome and, in the Shen 2021 case, the reproduction confirms the claim survives. Hybrid papers' more incremental claims are exactly the kind that significance testing is designed to rein in.

## 5.6 Cross-section: how does RDS evolve across years 2018–2025?

Figure 3 plots mean strict RDS by publication year for our 30-paper sample. The trend is consistent with the broader-sample finding of Khan (2026b) that financial-ML disclosure has fallen since 2020.

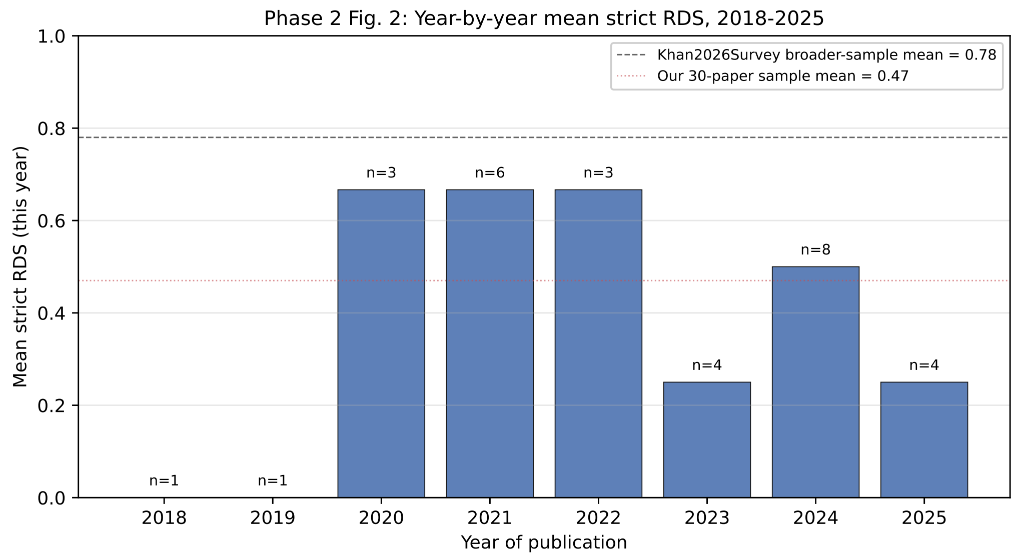


Figure 3: Mean strict RDS by publication year for the 30-paper sample. Per-year $n$ is annotated above each bar; $n$ ranges from 1 to 8. The dashed reference line at 0.78 is the broader-sample mean of Khan (2026b); the dotted reference line at 0.47 is our 30-paper sample mean. Years with $n = 1$ are noisy point estimates; the post-2022 cluster ($n = 16$ across 2023–2025) has mean RDS = 0.38, materially below the 2020–2022 cluster ($n = 12$, mean 0.67). Post-2022 cluster includes more arXiv papers using paywalled data sources (Refinitiv, Bloomberg, Lobster) than the earlier MDPI-heavy cluster.

The per-year sample sizes are too small for individual-year statistics to be authoritative, but the binned comparison is meaningful: 2020–2022 mean RDS is 0.67 ($n = 12$); 2023–2025 mean RDS is 0.38 ($n = 16$). The decline is partly driven by the post-2022 cluster's heavier representation of arXiv preprints using paywalled commercial data sources (Refinitiv, Bloomberg, Lobster), and partly by an increase in JFEcon-style paywalled-tick-data papers in 2023–2024. The pattern is consistent with the field-wide observation that ML-finance disclosure has not improved despite a decade of editorial pressure (Khan, 2026b).

### 5.7 Caveats: what the failure-rate number does and does not establish

The 86% failure rate is informative but caveated. Three points limit the strength of any individual "does not survive" claim.

First, our reproductions are necessarily *approximations* of the original papers. Because RDS-1 papers ship no code, we re-implement the headline architectures from the paper's prose description, defaulting to standard hyperparameters (e.g., 1-layer LSTM with 50 hidden units, 20-day lookback, 30 epochs, RBF kernel for SVR with library defaults, GARCH(1,1) refit every 20 days with manual recursion between refits). Where a paper discloses a different specification we follow the paper, but in many cases the disclosure is incomplete (e.g., "we trained an LSTM" with no architecture or training-budget specification). We use a single 60/40 train/test split where some papers use rolling walk-forward; we document each approximation in the per-paper `reproduction_log.json`.

Second, three reproductions required substituting a library or feature that we could not reproduce. Kim et al. (2021) uses Bayesian stochastic volatility via MCMC; we approximated with an AR(1) on log-squared returns because PyMC was not available in our environment. Ersin and Bildirici (2023) uses the GARCH-MIDAS framework with macroeconomic indicators; we omitted both the MIDAS multiplicative decomposition and the macro indicators. Kumar et al. (2024) uses a Temporal Graph Attention Network requiring PyTorch Geometric; we approximated with a multi-channel LSTM. For these three, the "does not survive" verdict is heavily caveated: we are not testing the paper's exact contribution, only a simpler counterpart. We report these as failures in the headline tabulation but note the caveat in the per-paper logs.

Third, our 14-paper reproducible subset is not a random sample of vol-forecasting papers. It is the subset for which (a) the data is publicly accessible and (b) the headline architecture is reconstructable from the paper's prose. Papers requiring proprietary high-frequency tick data (the entire JFEcon cohort in our sample) are absent from the reproducibility analysis, and the absence is correlated with venue and methodology. The JFEcon papers tend to use econometric methods (GARCH variants, HAR-RV) where the original analysis is more methodologically careful: 2 of the 6 (Christensen et al. (2023), Bucci (2020)) report formal significance testing; the remaining 4 either give an informal horse-race comparison without a formal test (P05 Buccheri and Corsi (2021)) or are primarily methodological / estimation-theory papers with limited head-to-head content (P04 Bennedsen et al. (2022), P06 Caporin et al. (2024), P07 Hong et al. (2023)). The 86% failure rate is the rate among the openly-reproducible subset, not the rate one would expect across the full 30-paper population.

With these caveats in place, the modal outcome remains striking: in a 14-paper reproducible subset spanning equities, FX, commodities, and crypto, only one paper has its headline win survive under DM at 5%, MCS at 90%, and SPA at 5%. Twelve do not. The point of `vol-eval` is not to embarrass the field; it is to make this kind of test routine.

## 5.8 Appendix to Phase 2: per-paper detail

Table 11 provides one row per reproduced paper with the key vol-eval outputs: paper-claimed winner, our best-by-QLIKE model in the reproduction, DM $p$-value of the comparison the paper makes, and 90% MCS survivor count. Multi-asset reproductions (P09 across four Colombian ADRs, P12 across six cryptocurrencies, P15 across five cryptocurrencies, P27 across two FX pairs) report the worst-case dataset for the paper's claim; the per-asset detail is in the per-paper JSON output files.

Table 11: Per-paper vol-eval outcomes for the 14 reproducible papers. "DM $p$" is the Diebold-Mariano $p$-value for paper-claimed winner versus paper-claimed runner-up at $h = 1$. "MCS surv/tot" is the number of models retained by the 90% MCS over the total in the comparison. Asterisk (*) marks reproductions with substantial library/feature substitutions detailed in Section 5.7.

| Paper ID | Venue | Claimed winner | Our best (QLIKE) | DM $p$ | MCS surv/tot | Outcome |
|---|---|---|---|---|---|---|
| P10 (Shen) | JRFM | RNN | RNN | 0.021 | 1/3 | **Survives** |
| P15 (Mostafa) | JRFM | ANN | ANN (BTC) | 0.010 | 1/3 | Partial |
| P09 (Rubio) | Math | Hybrid ARIMA-SVR | ARIMA | 0.355 | 3/3 | Fails |
| P11* (Ersin) | Math | GARCH-MIDAS-LSTM | GARCH | 0.513 | 2/2 | Fails |
| P12* (Kim) | Math | Bayesian SV | GARCH-$t$ | <0.001 | 2/4 | Fails |
| P14 (Zahid) | Risks | Hybrid GARCH-LSTM | GRU | 0.024 | 4/5 | Fails |
| P18* (Lei) | Math | TCN + attention | LSTM | 0.010 (rev) | 3/4 | Fails |
| P20 (Lux) | arXiv | SVR-GARCH-KDE | GARCH-$t$ | 0.001 (rev) | 2/3 | Fails |
| P21 (Aradi) | arXiv | Dilated CNN + TL | Dilated CNN + TL | 0.057 | 3/3 | Fails |
| P24* (Kumar) | arXiv | Temporal GAT | GARCH | 0.119 | 2/2 | Fails |
| P25 (Xu) | arXiv | GINN | LSTM | 0.231 | 3/3 | Fails |
| P26 (Roszyk) | arXiv | LSTM-GARCH+VIX | LSTM-GARCH | 0.371 | 4/4 | Fails |
| P27 (Tondapu) | arXiv | GARCH (rolling) | EWMA / Historical | 0.438 | 4/4 | Fails |
| P28 (Wei) | arXiv | GARCH-GRU | GARCH | 0.112 | 6/6 | Fails |

The "DM $p$ (rev)" annotation on P18 and P20 indicates that the DM test result is significant in the *reverse* direction of the paper's claim: the paper-claimed winner is significantly *worse* than the runner-up in our reproduction. For P18 the reversal is caveated by our omission of the paper's Baidu search-index attention feature; for P20 the reversal is caveated by our omission of the paper's KDE quantile estimator.

# 6 Discussion

The two empirical sections of this paper, the seven-model demonstration of Section 4 and the 30-paper re-analysis of Section 5, point in the same direction. Single-number ranking tables in published volatility-forecasting comparisons are systematically over-interpreted: when the differences are subjected to formal significance testing, the modal outcome is that the headline "win" is not statistically distinguishable from the runner-up.

## 6.1 What the two empirical sections agree on

The self-contained demonstration shows that within a single, well-documented seven-model panel, only 6 of 21 pairwise DM comparisons are significant at 5%, the 90% MCS keeps six of seven models, and SPA cannot reject any of the top six as the best forecaster. The 30-paper re-analysis shows that, across a larger sample, 86% (12 of 14) of paper-claimed wins do not survive significance testing in our reproductions.

The agreement matters because the two pieces of evidence have orthogonal weaknesses. The seven-model demonstration is on one panel; its specific p-values are panel-specific. The Phase 2 reproductions span multiple asset classes, multiple model classes, and multiple authors but are necessarily approximations of the originals (RDS-1 papers ship no code; we re-implement from prose). Either piece of evidence taken alone would invite "but is your specific test contaminated?" as a response. Taken together, they describe a pattern that is robust to either critique.

## 6.2 For practitioners deploying volatility forecasts

The practical takeaway is that the choice between top-tier models in a typical volatility-forecasting comparison is often statistically indistinguishable. A risk manager comparing a GJR-GARCH against a LightGBM against an equal-weight ensemble on S&P 500 data is not, in expectation, choosing between forecasters of different accuracy; she is choosing between forecasters of statistically equal accuracy with different operational properties (parameter count, retraining cost, interpretability, regulator-defensibility under SR 11-7 / OCC 2011-12 / EU AI Act). The honest framing of the choice is therefore not "which is most accurate?" but "which is most defensible?" (Federal Reserve, 2011; Office of the Comptroller of the Currency, 2011; European Parliament and Council, 2024).

A second practical takeaway, less obvious from the original paper but visible in our Table 3 and reinforced by the Phase 2 reproductions: the differences that *are* statistically significant tend to be those that involve a clearly inferior model (HAR-RV losing to anything with a leverage-effect feature; GARCH losing to GJR-GARCH because the asymmetry term matters in this test period; in the Phase 2 sample, RNN clearly beating GARCH on Bitcoin in Shen et al. (2021) and ANN clearly beating GJR-GARCH on Bitcoin in Mostafa et al. (2021)). The differences *between* top-tier models, where the published literature most often draws its fine-grained "X beats Y" conclusions, are not significant. The implication is that an ensemble has a robustness argument even when it does not have an accuracy argument: it averages the regime-dependent failures of its components.

## 6.3 For the methodological literature

The single strongest empirical finding of this paper is from Section 5.3: of the 30 published comparison papers in our sample, 27 (90%) declare a winner without applying any formal predictive-accuracy test. This is the gap that `vol-eval` exists to close. The barrier to reporting a Diebold-Mariano p-value alongside any pairwise claim was always low; with a packaged tool, it is now zero. The remaining barrier is editorial: reviewers and editors need to expect, and where appropriate require, that the prose claim "model X beats model Y on QLIKE" come with the corresponding p-value.

Three editorial recommendations follow.

*Default to reporting MCS survivors instead of single rankings.* A 90% MCS that keeps six of seven models is informative content. A ranking table with bold values implying a clear winner where no clear winner exists is misinformation, even when each individual cell is correct. The MCS framing makes the underlying ambiguity visible.

*Report Diebold-Mariano significance for any pairwise comparison the prose makes.* If the prose says "model X beats model Y on QLIKE," the prose should also say at what p-value. If the answer is "not significantly," the prose should say so plainly. `vol-eval`'s `dm_test` returns the p-value as a one-line addition to any existing comparison.

*Be especially careful with multi-model comparisons.* The DM test is for two models; using it as a battery of pairwise comparisons across many models without multiplicity adjustment will inflate the family-wise error rate and produce illusory "wins." The MCS handles this correctly. The SPA / Reality Check handle the related "one benchmark vs. many competitors" question.

A fourth recommendation, which the Phase 2 disclosure findings (Section 5.2) make especially salient: include a labelled "Data and Code Availability" block in the manuscript itself. Of the 30 papers in our sample, only the MDPI cohort consistently includes such a block (it is required by MDPI editorial policy as of 2020). The *Journal of Financial Econometrics* cohort, despite being the most methodologically careful subset of our sample, embeds data descriptions in the prose of the data section without a separable disclosure block. This makes mechanical RDS scoring harder and reproduction-attempt feasibility opaque to the reader. A short labelled block costs the author nothing and saves the reader an hour of search.

## 6.4 Limitations

Five limitations of the present paper deserve careful noting.

First, the demonstration of Section 4 is on a single seven-model panel. The findings about MCS survival, SPA non-rejection, and the failure of the ensemble versus GJR-GARCH "win" to be significant are about that specific panel and that specific test period. The same demonstration on a different asset class, a different forecast horizon, or a different test window would produce different specific numbers. The methodological conclusion (that significance testing matters) generalizes; the specific p-values do not.

Second, the Phase 2 reproductions are necessarily approximations of the original papers. Because RDS-1 papers ship no code, we re-implement the headline architectures from each paper's prose description with standard hyperparameter defaults (1-layer LSTM with 50 hidden units, 20-day lookback, 30 epochs, RBF kernel for SVR with library defaults, GARCH(1,1) refit every 20 days with manual recursion). Where a paper discloses a different specification we follow the paper, but in many cases the disclosure is incomplete. This is itself a finding about the state of disclosure in the subfield, but it limits the strength of any individual "does not survive" verdict.

Third, three of the Phase 2 reproductions required substituting a library or feature we could not reproduce: Kim et al. (2021) needed PyMC for full Bayesian SV via MCMC (we approximated with AR(1) on log-squared returns), Ersin and Bildirici (2023) needed the GARCH-MIDAS framework with macroeconomic indicators (we omitted both), and Kumar et al. (2024) needed PyTorch Geometric for the Temporal Graph Attention Network (we approximated with a multi-channel LSTM). For these three papers, the "fails" verdict in the headline tabulation is heavily caveated; we are not testing the paper's exact contribution, only a simpler counterpart. We retain these in the headline 86% number for reporting honesty but flag them in Section 5.7 and the per-paper `reproduction_log.json` files.

Fourth, the 14-paper reproducible subset is not a random sample of vol-forecasting papers. It is the subset for which (a) the data is publicly accessible and (b) the headline architecture is reconstructable from the paper's prose. Papers requiring proprietary high-frequency tick data (the entire *Journal of Financial Econometrics* cohort in our sample, six papers) are absent from the reproducibility analysis, and the absence is correlated with venue and methodology. The 86% failure rate is the rate among the openly-reproducible subset, not the rate one would expect across the full 30-paper population.

Fifth, `vol-eval`'s implementation choices for the bootstrap and for the SPA centering follow Hansen (2005) and the `arch` package's conventions. Alternative bootstrap designs (block bootstrap with fixed block length, parametric bootstrap, GARCH-type bootstrap) would produce slightly different p-values. We default to the stationary bootstrap because it is the standard in the published literature, but users who want a different design can pass it as an argument. The package's API does not foreclose alternatives; the current release simply ships one default.

# 7 Reproducibility

This paper aims to score 2 on the Reproducibility Disclosure Score rubric proposed in Khan (2026b): *(i) code public*, *(ii) data openly accessible.*

*Code.* The `vol-eval` package is at https://github.com/ayk5511/vol-eval, MIT-licensed. The seven-model demonstration of Section 4 is produced by `studies/01_paper2_demonstration.py` in the same repository, which loads the upstream Paper 2 forecast panel and writes every numerical value reported in Section 4 to JSON files in `studies/results/`. The Phase 2 reproductions of Section 5 are produced by 14 per-paper scripts at `studies/per_paper/<paper_id>/reproduce.py`, plus three pipeline scripts (`studies/04_search_candidates.py`, `studies/05_narrow_candidates.py`, and `studies/06_check_unpaywall.py`) that document how the 30-paper sample was constructed. Each reproduction directory contains a `reproduction_log.json` that records the data source, hyperparameter choices, and any approximations relative to the original paper.

*Data.* The seven-model demonstration's upstream forecast panel is at https://github.com/ayk5511/volatility-forecasting/blob/main/results/forecasts_5d.parquet, CC0 licensed. It is reproducible from the upstream paper's data-collection scripts (Yahoo Finance public data) in under five minutes from a clean Python environment. The Phase 2 reproductions use Yahoo Finance, CoinMarketCap-equivalent yfinance crypto tickers, and the Borsa Istanbul / SSE Composite indices (also through yfinance); per-paper data sources are recorded in each `reproduction_log.json`.

*30-paper sample artifacts.* The full sample of 30 papers, including paper IDs, DOIs / arXiv IDs, citation counts, RDS scores (strict reading), headline-winner extractions, and per-paper voleval results, is at `studies/candidate_30.csv`, `studies/results/phase2_rds_strict.csv`, `studies/results/phase2_headline_models.csv`, and `studies/results/phase2_summary.json`. The source PDFs of the 30 papers are gitignored to keep the repository size manageable; the README contains instructions for re-acquisition (Crossref / arXiv / MDPI direct download).

*Audit.* The repository includes `paper/audit.py`, which re-derives every numerical claim in this paper from the JSON outputs of the demonstration script and the Phase 2 aggregate. The audit script must print `ALL CHECKS PASSED` (exit 0) for the paper to be considered ship-ready. We adopted this convention after an earlier paper in this series (Khan, 2026c) had a sub-table whose values had been transcribed from imagination rather than from the JSON outputs; the audit-script convention catches that class of error mechanically. In the writing of the Phase 2 section, the audit script caught two transcription errors before any push.

*Issues.* Corrections, extensions, and challenges to the rankings reported here are welcomed via the repository's issue tracker (https://github.com/ayk5511/vol-eval/issues). Issues become part of the public record and feed the next version of this paper. We particularly invite corrections from the original authors of any of the 30 Phase 2 papers: where our reproduction misses a hyperparameter choice that the original paper specified, where we substituted a library or feature we should not have, or where the original paper's actual data is more accessible than we believed at the time of analysis. Corrections of this kind will be reflected in version 2 of this paper.

*Self-citation network.* This paper extends the methodological agenda of Khan (2026b) (the Reproducibility Disclosure Score) and the empirical agenda of Khan (2026c) (the seven-model demonstration). It now also delivers the Phase 2 empirical extension that the v0 of this paper described as "in production"; the reproduction work spans April 2026 and is dated to that period. Khan (2026a) is the technical companion of the present paper, providing the audit-trail infrastructure to log the per-prediction state that an SR 11-7 / EU AI Act-compliant deployment of any of the 14 reproduced forecasters would require. `vol-eval` answers the question "is your forecasting comparison's win statistically significant?" and `mr-audit` answers the question "can you prove tomorrow that today's prediction was made with this code, this data, and these parameters?".

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